POWER SUPPLY: 9333063

This specification is approved for use by the U.S. Army Armanent Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1. 1 Scope. This specification covers Power Supply 9333063, part of Radar Test Set AN/TPM-2 2A. The power supply provides central supply and control of ac and regulated dc voltage required for operation of Radar Test Set AN/T PM-22A and reference voltages for test equipment in Test Set Group, AA Gun Mount, AN/TSM-100A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of issue listed in that issue of the Department of Defense Index of specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Amy Materiel; Packaging and Packing of

FSC 4931

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Armament Research and Development Center Attn: DRSMC-QA Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MIL-C-46357	- Computer, Gun Direction; M18 Manufacturing Process and Inspection of
MIL-B-48594	- Board Assembly, Component 11748254
MIL-A-50300	- Amplifier, Analog Voltage: 10554887
MIL-C-50329	- Converter, Electronic: 10549013

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

F9333063	- Power Supply, PP-6324
F9333085	- Schematic, Power Supply

PACKAGING DATA SHEET

F9333063	- Packaging of Power Supply

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Fabrication. The power supply shall be manufactured in accordance with Drawing F9333063 and drawings pertaining thereto. The following subassemblies shall have been inspected and accepted in accordance with their respective drawings and specifications prior to assembly in the power supply.

Unit No.	Name	Specification
A1	Converter, Electronic: 10549013	MIL-C-50329
A3	Amplifier, Analog Voltage: 10554887	MIL-A-50300
A4	Board Assembly, Component: 11748254	MIL-B-48594

3.2 General specifications. Where applicable, the power supply shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishings

d. Part identification and marking

e. Workmanship

f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies

b. Wiring

c. Threaded fasteners

d. Marking

e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The power supply shall be mounted on the vibration table in normal service position. The power supply

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shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/see). Vibration tests shall be conducted under ambient room-conditions.

3.3.2 Shock. The power supply shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating. The power supply shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-operating. The power supply shall meet the requirements of 3 4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The power supply shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The power supply shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours, and +160°F for a period of 48 hours (after temperature stabilization). Power supply shall also meet the requirements of 3.4 and 3.5 when returned to room ambient temperature.

3.3.5 Humidity (non-operating). The power supply shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95-percent for a period of 10 days.

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3.4 Electrical.

3.4.1 Continuity. continuity and isolation of conductors shall be as specified on the schematic diagram 9333005.

3.4.2 Circuit breakers. Each circuit breaker (CB1 through CB4) listed below shall be closed when reset pushbutton is depressed so white band is concealed. Each circuit breaker shall be open when reset pushbutton is pulled out (manually or by overload) to expose white band.

Panel Marking	Reference Designation
115 V - 400 Hz (2A)	CB1
PWR SUP 1 (35A)	CB2
TESTER CKT (10A)	CB3
CONTROL CKT (5A)	CB4

3.5 Performance. To meet the following requirements, the subassemblies listed in 3.1 shall be installed in the power supply.

3.5.1 115V, 400 Hz ON-OFF switch S1. With 115 ± 10 volts, 400 Hz power connected to J1-A and B, +24 volts dc power connected to J1-D(+) and E(RET), TEST MODE switch S2 in any position except FC CONSOLE, terminals J2-K and S shorted, and circuit breakers CB2 and CB4 closed, 115V, 400 Hz switch S1 shall serve to change indicator DS1 from OFF to ON and the reverse, and to cause 115 ± 10 volts, 400 Hz power to be applied to J2-A and L in accordance with indicator DS1.

3.5.2 +24 Vdc ON-OFF switch S3. With +24 volts dc power connected to J1-D(+) and E(RET), TEST MODE switch S2 in any position except FC CONSOLE, terminals J2-K and S shorted, and circuit breakers CB2 and CB4 closed, +24 Vdc switch S3 shall serve to change indicator DS3 from both ON and OFF illuminated to only ON illuminated and the reverse, and to cause +24 volts dc power to be applied to points specified on the schematic drawing, in accordance with indicator DS3. Indicator 115 V, 400 cps (DS1) shall track DS3 indicator.

3.5.3 Power SUP 1 voltages. With power applied as in 3.5.2, CB2 and CB4 closed, S3 ON, TEST MODE switch S2 in any position except FC CONSOLE, terminals J2-K and S shorted, VOLTAGE selector switch S4 shall serve to register voltages of converter on VOLTS meter M1, in accordance with switch positions and indicated voltages of Table I.

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TABLE I. Converter voltages.

POSITION OF VOLTAGE SWITCH (S4)	READING ON VOLTS METER (Ml)
OFF	0 ± 1
+24V	+24 ± 2
+6	+6 ± 1
+15	+15 ± 1
-15	-15 ± 1
+23	+23 ± 1
-23	-23 ± 1
+29	+29 ± 1
+33	+33 ± 1
-33	-33 ± 1

3.5.4 Dc voltage outputs. With power applied as in 3.5.2, CB2, CB3, CB4 closed, S3 ON, J2-K and S shorted, and TEST MODE switch S2 in any position except FC CONSOLE, the voltages shall appear at the terminals as indicated in Table II.

TABLE II. Output voltages.

CONNECTOR TERMINALS	NO LOAD DC VOLTS
J3-P(+) to J3-R(RET)	+24 ± 2
J3-S(+) to J3-T(RET)	+24 ± 2
J3-A(+) to J3-B(RET)	+15 ± 1
J3-E(-) to K3-F(RET)	-15 ± 1
J3-C(+) to J3-H(RET)	+23 ± 1
J3-c(-) to J3-W(RET)	-23 ± 1
J3-h(+) to J3-d(RET)	+33 ± 1
J3-j(-) to J3-e(RET)	-33 ± 1

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TABLE II. Output voltages - continued

CONNECTOR TERMINALS	NO LOAD DC VOLTS
J3-U(+) to J3-O(RET)	+29 ± 1
J3-k(+) to J3-f(RET)	+ 6 ± 0.5
J4-M(+) to J4-W(RET)	+33 ± 1
J4-N(-) to J4-A(RET)	-33 ± 1
J4-B(+) to J4-C(RET)	+29 ± 1
J4-E(+) to J4-G(RET)	+ 5 ± 0.5
With S2 in FC Position	
J4-F(+) to J4-G(RET)	+ 5 ± 0.5

3.5.5 Power control voltage. With power applied as in 3.5.2, CB4 closed, TEST MODE switch S2 shall serve, when set in turn to switch positions shown in Table III, to apply a 24 volt power control signal between indicated terminal of J2 and +24 volts return at J1-E.

TABLE III. Power control voltages.

POSITION OF TEST MODE SWITCH (S2)	J2 TERMINAL CARRYING 24 VOLT POWER CONTROL VOLTAGE
PWR/SUP SERVO	N
MICORWAVE/KLYSTRON PWR SUP	E, D
RCVR/XMTR	D, P
RCVR	F, P
RANGE COMPUTER	R, F
PC CARDS	P

3.5.6 FC control voltages. With power applied as in 3.5.2, CB4 closed, TEST MODE switch S2 in FSC CONSOLE position, the voltages of Table IV shall appear at the terminals indicated in Table IV.

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TABLE IV. FC control voltages

TEST POINTS	J4 TERMINALS	DC VOLTS NO LOAD
+10(TP3) to COM(TP6)	J4-X(+) to J4-S(COM)	+10 ± 0.l
-10(TP4) to COM(TP6)	J4-U(-) to J4-S(COM)	-10 ± 0.1
+20(TP5) to COM(TP6)	J4-T(+) to J4-S(COM)	+20 ± 0.2

3.5.7 10 KV Voltage divider. With +24 Vdc power connected to J1-D(+) and J1-E(RET), CB4 closed, TEST MODE switch, S2, placed in MICROWAVE/KLYSTRON PWR SUP position, and 10 KV DC applied to SJ1, a voltage of 9.1 Vdc ± 5% shall be present at TP9 (+) and TP10 (-). When switch S6 is depressed this voltage shall fall to 0 ± 0.4 Vdc, With S6 released and S5 depressed the high voltage power supply output current shall increase by 100MA ± 10%.

3.5.8 HI-Voltage interlock. Resistance between TP10, TP11, TP12, and TP13 shall be less than 2.5 ohms.

3.6 First article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.7 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Unless otherwise specified in the item detail specification, or contract, the following types of inspection shall be conducted on the unit or product:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

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4.3 First article inspection.

4.3. Submission. The contractor shall submit a first article as designated by the contracting officer (see 6.2) for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 3 test sets.

4.3.2 Inspection to be performed. First article samples may be subjected by the Government to any or all of the examinations and tests specified in 4.4.2 of this specification and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units Of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisiona, same specifications and the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless Of the type of inspection procedure which is being applied to determine conformance with requirements. Definition of inspection terms shall be as listed in MIL-STD-109.

4.4.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected. in accordance with the “Inspection provisions for components and subassemblies” and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables V and VI and methods of inspection in 4.7. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. Workmanship shall be in accordance with MIL-STD-454 and other applicable specifications.

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TABLE V. Classification of defects.

100% Inspection.

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Continuity	3.4.1	4.7.2.1
102.	Circuit breakers	3.4.2	4.7.2.2
103.	115V, 400 Hz ON-OFF switch	3.5.1	4.7.3.1
104.	+24 Vdc ON-OFF switch	3.5.2	4.7.3.2
105.	Converter voltages and monitoring	3.5.3	4.7.3.3
106.	DC voltage outputs	3.5.4	4.7.3.4
107.	Power control voltage	3.5.5	4.7.3.5
108.	FC control voltages	3.5.6	4.7.3.6
109.	10KV Voltage divider	3.5.7	4.7.3.7
110.	HI-Voltage interlock	3.5.8	4.7.3.8

4.4.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Special sampling. One power supply out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample. shall have met the requirements and tests in Table V and shall then meet the requirements and tests in Table VI.

TABLE VI. Special sampling.

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General Specifications	3.2	MIL-F-13926/ MIL-C-46357 visual

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TABLE VI. Special sampling - continued

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Vibration	3.3.1	4.7.1.1
304. Shock	3.3.2	4.7.1.2
305. Altitude (operating)	3.3.3.1	4.7.1.3
306. Altitude (non-operating)	3.3.3.2	4.7.1.3
307. Temperature	3.3.4	4.7.1.4
308. Humidity (non-operating)	3.3.5	4.7.1.5

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved. (see 4.4.3).

4.6 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the referenced specifications and standards cited in 4,4.2. Inspection equipment shall be listed as in Table VII. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See 6.3 of this specification.

TABLE VII. Inspection equipment list.

Item No.	Item	Description
1.	Power supply, ac	115 ± 10 volts, ac, 400 Hz, single phase, 10 amperes
2.	Power supply, dc	+24 ± 2 volts dc, 40 amperes, (negative side grounded)
3.	Voltmeter, digital	0 - 200 volts ac, 0-50 volts dc, accuracies of ± 2% on dc and ± 3% on ac. Sensitivities of 20,000 ohms per volt dc and 5000 ohms per volt ac
4.	Power supply	0 to 15KV DC at 300MA; Accuracy = ± 1%
5.	Ohmmeter	Accuracy and Resolution of 0.2% or better from 0 to 1000 Kilohms

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Procedure I of MIL-STD-810 as modified by paragraph 3.3.1 herein. After vibration, the power supply shall be tested as specified in the subparagraphs of 4.7.2 and 4.7.3 for compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulse shall be 11 ± 1 ms and 15 g's. The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the power supply shall be tested as specified in 4.7.2 and 4.7.3 for compliance with 3.3,2.

4.7.1.3 Altitude (non-operating and operating. This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810 . The operational checkout shall be as specified in 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501, of MIL-STD-810, with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502, of MIL-STD-810, with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with procedure III, Method 507 of MIL-STD-810 except that in Step 7, the operational checkout shall be performed at the end of the tenth cycle only, and at standard ambient conditions.

4.7.2 Electrical.

4.7.2.1 Continuity. Continuity shall be inspected by standard

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measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.7.2.2 Circuit breakers. Circuit breakers CB1 through CB4 shall be inspected in accordance with the following procedures for compliance with 3.4.2.

4.7.2.2.1 115V, 400 Hz circuit breaker (CB1). Connect J2-K to J2-S. Put TEST MODE switch S2 in any position except FC CONSOLE. Close 115V, 400 Hz circuit breaker CB1 and 24 Vdc CONTROL CKT circuit. breaker CB4 by depressing their pushbuttons so white bands are concealed. Connect 115 volts ac, 400 Hz, and +24 volts dc input power from power supplies, items 1 and 2, to connector J1 in accordance with Figure 1. Either ON or OFF legend of 115V, 400 Hz pushbutton indicator switch (S1) shall light. If OFF legend lights, depress pushbutton and OFF legend shall be extinguished and ON legend shall light. Connect voltmeter, item 3, across ACVM test points (TP1 and TP2). Voltage shall be 115 volts rms, nominal. Open 115V, 400 Hz circuit breaker by pulling out reset button so white band is exposed. Voltmeter shall read 0 and ON legend of 115V, 400 Hz pushbutton indicator shall be extinguished. Disconnect ac input power, close l15V, 400 Hz circuit breaker by depressing pushbutton, and again connect ac input power. The ON legend of 115V, 400 Hz pushbutton indicator shall light and voltmeter shall read 115 volts rms, nominal.

4.7.2.2.2 PWR SUP 1 circuit breaker (CB2). Connect J2-K to J2-S . Put TEST MODE switch S2 in any position except FC CONSOLE. Close PWR SUP 1 circuit breaker CB2 and 24 Vdc CONTROL CKT circuit breaker CB4 by depressing their pushbuttons so white bands are concealed. Connect +24 volts dc input power from power supply, item 2, to connector J1 in accordance with Figure 1. Either ON or OFF legend indicator DS3 associated with +24 Vdc switch S3 shall light. If OFF legend lights, depress switch S3 and OFF legend shall be extinguished and ON legend shall light. Turn VOLTAGE switch S4 to +24V and VOLTS meter M1 shall read +24 ±2 volts dc. Open PWR SUP 1 circuit breaker by pulling out reset button so white band is exposed. VOLTS on meter Ml shall read 0 and ON legend of ±24 Vdc pushbutton indicator shall be extinguished. Disconnect dc input power, close PWR SUP 1 circuit breaker by depressing pushbutton and again connect DC input power. The ON legend of +24 Vdc pushbutton indicator shall light and VOLTS on meter Ml shall read +24 ±2 volts dc.

4.7.2.2.3 TESTER CKT circuit breaker (CB3). Connect J2-K to J2-S. Put TEST MODE switch S2 in any position except FC CONSOLE. Close PWR SUP 1 circuit breaker CB2 and 24 Vdc CONTROL CKT circuit breaker CB4 by depressing their pushbuttons so white bands are concealed. Close TESTER CKT circuit breaker CB3 by depressing pushbutton so white band is concealed Connect +24 volts dc

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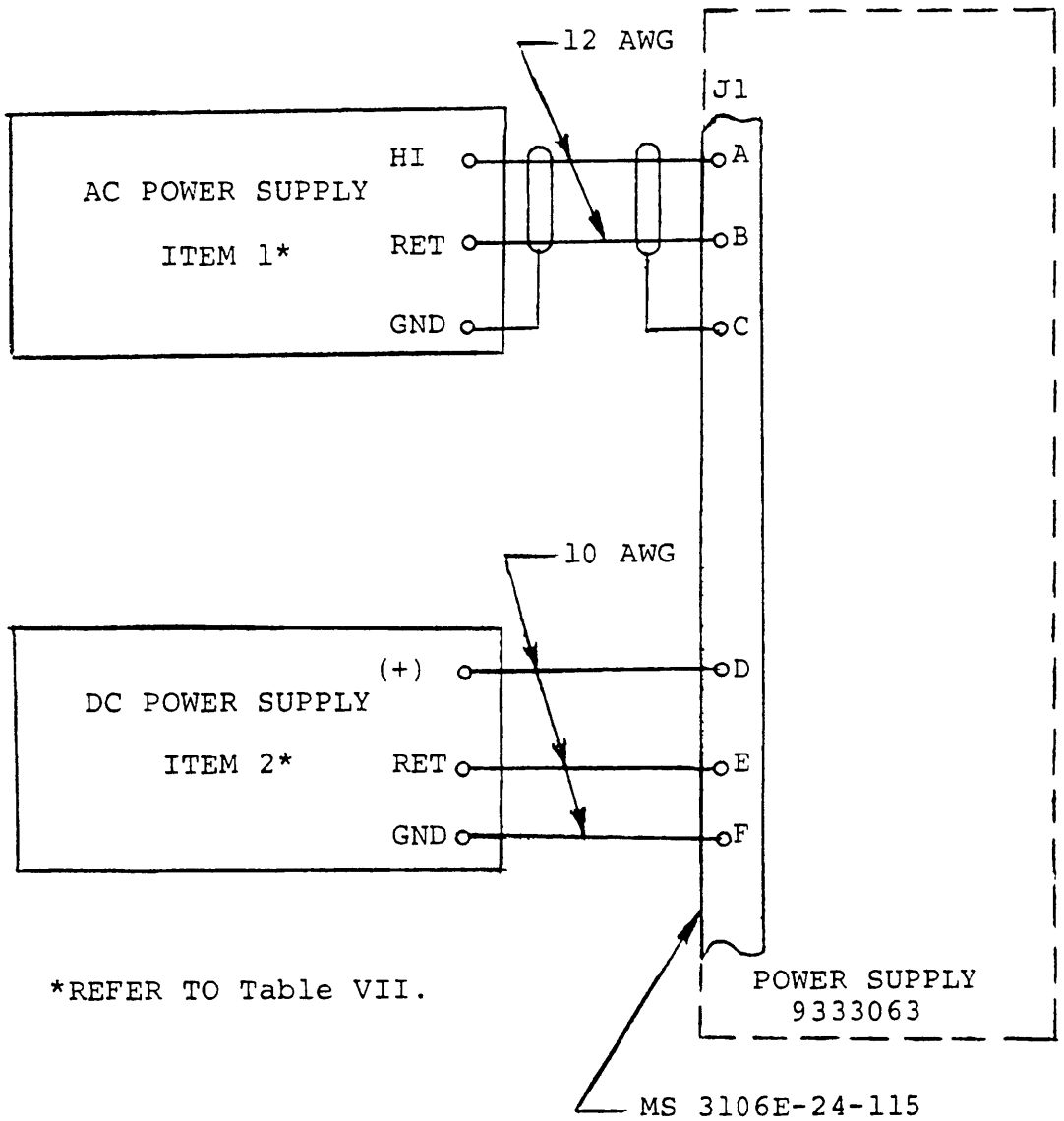


Figure 1. Power input connections.

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input power from power supply, item 2, to connector J1 in accordance with Figure 1. Push +24 Vdc switch S3 to ON (ON legend of DS3 illuminated). Connect voltmeter, item 3, between J3-P (+) and J3-R (RET). Voltage shall be +24 ±2 volts dc. Open TESTER CKT circuit breaker by pulling out. reset button so white band is exposed. Voltmeter shall read 0. Disconnect dc input power, close TESTER CKT circuit breaker by depressing pushbutton, and again connect dc input power. The voltmeter shall read +24 ±2 volts dc.

4.7.2.2.4 CONTROL CKT circuit breaker (CB4). Connect J2-K to J2-S. Turn TEST MODE switch S2 to the PWR SUP/SERVO position. Close CONTROL CKT circuit breaker CB4 by depressing pushbutton so white band is concealed. Connect +24 volts dc input power from power supply, item 2, to connector J1 in accordance with Figure 1. If indicator DS3 reads OFF', push switch S3 so that DS3 reads ON. Connect voltmeter, item 3, between J2-N (+) and J2-C (RET). Voltage shall be +24 ±2 volts dc. Open CONTROL CKT circuit breaker by pulling out reset button so white band is exposed. Voltmeter shall read 0. ON legend of DS3 shall be extinguished. Disconnect dc input power, close TESTER CKT circuit breaker by depressing pushbutton and again connect dc input power. Voltmeter shall read +24 ±2 volts dc and ON legend of DS3 shall light.

4.7.3 Performance. Turn circuit breakers and switches as listed below prior to performing any test procedures.

Circuit Breakers	Reference Designation	Position
115V 400 Hz (10A)	CB1	Open (white band exposed)
PWR SUP 1 (35A)	CB2	Open (white band exposed)
TESTER CKT (10A)	CB3	Open (white band exposed)
CONTROL CKT (2A)	CB4	Open (white band exposed)

Switches	Reference Designation	Position
TEST MODE rotary switch	S2	OFF (full CCW)
VOLTAGE rotary switch	S4	OFF (full CCW)

4.7.3.1 115V, 400 Hz ON-OFF switch S1. Connect input power from power supplies, items 1 and 2, to connector J1 in accordance with Figure 1. Put TEST MODE switch S2 in any position except FC CONSOLE. Connect J2-K to J2-S. Close 115V, 400 Hz circuit breaker CB1 and CONTROL CKT circuit breaker CB4 by depressing

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their pushbuttons so white bands are concealed. Either ON or OFF legend of 115V, 400 Hz indicator DS1 shall light. If OFF legend lights, depress 115V, 400 Hz switch S1 and OFF legend shall be extinguished and ON legend shall be illuminated. Connect voltmeter, item 3, across ACVM test points (TP1 and TP2) and J2-A and J2-L Voltage shall be 115 volts rms, nominal, when 115V, 400 Hz switch is ON, and zero when 115V, 400 Hz switch is OFF. Operate 115V, 400 Hz switch to ON-OFF three times, leave in OFF position.

4.7.3.2 +24 Vdc ON-OFF switch S3. Connect +24 volts dc from power supply, item to J1 in accordance with Figure 1. Put TEST MODE switch S2 in any position except FC CONSOLE. Connect J2-K to J2-S. Close PWR SUP 1 and CONTROL CKT circuit breakers (CB2 and CB4) by depressing pushbuttons so white bands are concealed. Either ON or OFF legend of +24 Vdc indicator DS3 shall light. If OFF legend lights, depress switch S3 and OFF legend shall be extinguished and ON legend shall light. Turn VOLTAGE switch S4 to +24V and VOLTS meter M1 shall read +24 ±2 volts dc. Depress +24 Vdc ON-OFF switch. ON legend shall be extinguished and OFF legend shall light and VOLTS meter Ml shall read 0. Operate switch to ON-OFF three times. Leave indicator DS3 reading ON. Observe indicator 115V, 400 cps DS1 tracks DS3 indicator.

4.7.3.3 Converter voltages and metering. After completing 4.7.3.2, turn VOLTAGE selector switch S4 to each of its 10 Positions and at each position the voltage reading given in Table I shall register on VOLTS meter Ml in compliance with 3.5.3.

4.7.3.4 Dc output voltages. With input power applied as in 4,7.3.1, CB2, CB3, and CB4 closed, DS3 reading ON, J2-K and S shorted, and TEST MODE Switch S2 in any position except FC CONSOLE, use digital voltmeter, item 3, to measure polarity and amplitude of dc no-load output voltages at J3 and J4 terminals given in Table II for compliance with 3.5.4.

4.7.3.5 Power control voltage. Close CONTROL CKT circuit breaker CB4. Connect power as in 4.7.3.1. Use voltmeter, item 3, to measure 24 ±2 volts dc from each terminal specified in Table III to J1-E, as TEST MODE switch S2 is set to the position specified in Table III.

4.7.3.6 FC control voltages. Turn TEST MODE switch S2 to FC CONSOLE. Close CONTROL CKT circuit breaker CB4. Apply power as in 4.7.3.2. Use voltmeter, item 3, to measure voltage at test points and terminals given in Table IV for compliance with 3.5.6.

4.7.3.7 10 KV Voltage divider. Connect +24 Vdc from power supply, item 2, to J1 in accordance with Figure 1. Place TEST

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MODE switch S2 in MICROWAVE/KLYSTRON PWR SUP position, and close CB4 by depressing pushbutton 50 white band is concealed. With power supply, item 4 of Table VII, apply 10KV to high voltage terminal SJ1. With digital voltmeter, item 3 of Table VII, verify the presence of 9.1V at TP9 (+) and TP10 (-) per the requirements of 3.5.7 Depress S6, and verify that the 9.1V fall to 0 ± 0.4 Vdc per the requirements of 3.5.7. Release S6, depress S5, and verify the increase in high voltage current per 3.5.7.

4.7.3.8 HI-Voltage interlock. With an ohmmeter, item 5 of Table VII, verify the resistance requirements of 3.5.8 between the test points indicated.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 9333063.

6. NOTES

6.1 Intended use. The power supply provides central supply and. control of ac and regulated dc voltages required for operation of Radar Test Set AN/TPM-22A(SMSS). The power supply contains 1 dc/dc converter an analog voltage amplifier and a high voltage load circuit. The de/de converter operates from +24 volts dc and provides regulated dc voltages of +15, -15, +23, -23, +33, -33, +29 and +6 for the SMSS test equipment. The analog voltage amplifier provides voltages of +10, -10 and +20 for the test equipment in Test Set Group, AA Gun Mount, AN/TSM-100A. The high voltage load circuit provides switching and loading for the Klystron Beam Power Supply.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

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f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment of product.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, tailored, with appropriate addendum to each. See 6.3.

b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing, as defined in DOD-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which Government designs are not specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor calibration procedures and mainten-

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ance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number and defect number from Section 4 shall

be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged-. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAF-I (D), Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research and Development Command, (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-50346A	POWER SUPPLY, 9333063

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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